

# Probabilistic Prediction of Placements in Teamfight Tactics: A Bayesian Nonparametric Approach with BART

Alain Phillip Jay

Mathematical Engineering, Universidad EAFIT

Medellín, Colombia

ajayr@eafit.edu.co

June 25, 2026

## Abstract

This work aims to predict dynamic survival probabilities and relative performance in the competitive video game Teamfight Tactics (TFT) from players' recent histories and the competitive context of the lobby. We formulate the problem as a discrete-time survival analysis, where time is measured by the final round reached by each player. To overcome the limitations of parametric proportional-hazards models, we implement a Bayesian nonparametric approach using Bayesian Additive Regression Trees (BART). Through longitudinal data expansion and the use of latent variables with a probit link, the model estimates elimination probabilities while implicitly capturing complex in-game interactions. Trained exclusively on pre-match information, the selected model recovers the relative ranking of players within a lobby (C-index = 0.630) and reproduces the staged dynamics of real matches; all observed eliminations fall within the model's 90% and 95% credible intervals. However, the width of these intervals indicates that pre-match signal predicts relative merit ordering well but is insufficient to isolate the exact round of elimination, reflecting the irreducible in-game variance of TFT.

## 1 Introduction

Survival analysis traditionally addresses the time until the occurrence of an event, modeling the hazard function and the survival probability  $S(t)$ . In the context of the video game Teamfight Tactics (TFT), where eight players compete until their health points reach zero, the duration of a match can be naturally formulated as a survival time. Specifically, the discrete time scale in this work is defined using the final round reached

---

This project was developed under Riot Games' "Legal Jibber Jabber" policy using assets owned by Riot Games. Riot Games does not endorse or sponsor this project. *Teamfight Tactics* and *Riot Games* are trademarks or registered trademarks of Riot Games, Inc.

by each player before being eliminated. Historically, semiparametric approaches such as the Cox proportional-hazards model have been dominant for relating covariates to survival time. In practice, however, the variables of interest often exhibit non-proportional effects and highly nonlinear relationships. Recent models based on ensembles of regression trees, in particular BART (Bayesian Additive Regression Trees), offer an extremely flexible framework that outperforms traditional competitors without imposing restrictive functional forms, and provides a natural quantification of uncertainty. This project applies that methodology with the goal of predicting dynamic survival probabilities and relative performance in TFT from players’ recent histories and the competitive context of the lobby. In this way, the model does not use any information from the target match as a predictor; instead, it bases its estimates solely on historical information available prior to the match and on the relative characteristics of the opponents.

## 2 Methodology

### 2.1 Construction of the Empirical Dataset and Filters

The empirical database was built by extracting match histories using the Riot Games API, resolving player identities through their PUUID. To ensure the homogeneity and competitive maturity of the data, the analytical cohort was restricted to standard ranked matches played on the recent patches 16.4, 16.5, and 16.6. Analytical filters were applied to exclude atypical or early-abandonment matches, requiring that players reach at least level 4 and survive beyond round 18.

A central element of the methodological design was the application of a per-player **burn-in** period: the first 10 observed matches of each subject were used exclusively to build their statistical history and stabilize their prior metrics, but were not included as target observations in model training.

### 2.2 Data Structure and Survival Time

To implement the BART algorithm in a survival problem, we transform the approach into one of sequential binary classification over discrete time. In this empirical application, the time scale is operationalized strictly through the final round reached by the player. Although match duration can be grouped into stages, these are used only as descriptive variables or auxiliary covariates, since modeling directly over the round preserves greater temporal resolution.

Each record is expanded longitudinally over this grid of rounds. The elimination event (**status** = 1) is assigned to players who finish between 2nd and 8th place, whereas the winner experiences right-censoring (**status** = 0), ending their participation with a sequence consisting exclusively of zeros.

### 2.3 Historical Covariates, Lobby Context, and Smoothing

The feature vector  $x_i$  feeding the model omits any information observed during the target match, ensuring that **predictions are based exclusively on prior information**. The constructed covariates fall into two main categories:

- **Historical player attributes:** Computed solely from the past available up to the current match. They include the number of previous matches (experience), and historical averages of placement, final round, Top-4 rate, win rate (`win_rate`), final level, and remaining gold. To avoid unstable estimates for players with short histories, these metrics were stabilized through empirical-Bayes shrinkage, pulling individual averages toward the global average. Historical standard deviations were also included to capture player consistency.
- **Lobby-relative attributes:** The competitive context of the matchup was reconstructed by incorporating relative metrics against the available opponents. These covariates assess relative strength and include direct advantages over the lobby (e.g., `placement_adv_vs_lobby`, `survival_adv_vs_lobby`), relative consistency, historical rankings within the lobby, and the count of empirically stronger opponents.

## 2.4 Probit Model and Latent Variables

To connect the discrete sequence of zeros and ones with BART’s continuous regression model, we resort to the data-augmentation methodology of Albert and Chib. We introduce an unobservable continuous latent variable  $z_{ij} \sim N(\mu_{ij}, 1)$ , which we condition on the observed data: it is truncated to the interval  $(0, \infty)$  if the player was eliminated ( $y_{ij} = 1$ ), and to  $(-\infty, 0)$  if they survived the stage ( $y_{ij} = 0$ ).

The theoretical mean of this latent variable is dictated by the model’s predictor:

$$\mu_{ij} = \mu_0 + f(t_{(j)}, x_i) \quad (1)$$

where  $\mu_0$  is a baseline temporal centering parameter and  $f(t_{(j)}, x_i)$  represents the continuous sum collected from the BART trees, which simultaneously evaluate the game stage  $t_{(j)}$  and the player feature vector  $x_i$ , composed of the historical and lobby-relative variables described in the previous section.

Finally, the predictive elimination probability is estimated by applying the probit link function:  $p_{ij} = \Phi(\mu_{ij})$ , where  $\Phi(\cdot)$  denotes the Cumulative Distribution Function (CDF) of the Standard Normal distribution.

## 2.5 Prior Regularization and Weak Learners

BART’s robustness against overfitting stems from highly restrictive prior distributions that force each tree to be a weak learner.

- **Tree structure:** The probability that an interior node splits decays strongly with depth  $d$  according to  $\alpha(1 + d)^{-\gamma}$ . We adopt the standard hyperparameters  $\alpha = 0.95$  and  $\gamma = 2$  to force shallow trees.
- **Terminal nodes (shrinkage):** The predictive contributions  $\mu_{jk}$  of each terminal leaf are modeled as  $\mu_{jk} \sim N(0, 2.25/m)$ , where  $m = 200$  is the number of trees. This ensures that individual predictions shrink toward zero.

## 2.6 Computational Implementation (MCMC)

Since the joint posterior distribution cannot be solved analytically, estimation was carried out using Markov Chain Monte Carlo (MCMC). The computational implementation of this project was carried out in the R environment. The main algorithmic and data-engineering effort consisted of extraction via the API, the stabilization of player histories (burn-in), and the restructuring and longitudinal expansion of the survival data into the required sequential binary format. Once the matrix was expanded, model fitting leveraged the Albert–Chib probit latent-variable architecture, using specialized binary-BART packages already available in R to run the posterior sampling of the tree topology and their terminal nodes.

## 3 Results

As a preliminary step before inference on the Teamfight Tactics tournament records, model evaluation was divided into two stages: a methodological validation on simulated scenarios and a pilot empirical test on real game data.

### 3.1 Methodological Validation on Simulated Scenarios

Before processing the empirical records, a pilot computational validation was run by simulating the controlled scenarios proposed by Sparapani et al. (2016).

In the base single-sample scenario, designed to recover a true Weibull distribution with a target censoring rate of 20%, the predictions of BART’s posterior mean survival function tracked the theoretical curve reasonably well, systematically outperforming the traditional Kaplan–Meier estimator in terms of Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

In the two-group scenario with crossing survival curves (non-proportional hazards), where traditional parametric methods suffer empirical degradation, the BART model successfully captured the diverging trajectories, demonstrating consistent improvements and reducing both RMSE and MAE relative to independent estimates. The pilot coverage probabilities of the Bayesian intervals were high and robust, confirming the mathematical viability of the architecture.

Once the methodological validation on simulated scenarios was passed, the model was applied to the empirical records of the Teamfight Tactics tournament. The central empirical objective was to evaluate the predictive capacity of the algorithm using only the players’ historical information and the relative context prior to the target match.

### 3.2 Training and Selection of Empirical Models

For the empirical modeling phase, an experimental design was structured around the training and comparison of four distinct specifications of the BART model. These variations arose from the combination of two data-processing approaches and two levels of covariate dimensionality:

1. **Training datasets:** Two main datasets were used. The first corresponded to the original dataset extracted directly through the Riot Games API, characterized

by partial lobbies (approximately 4 reconstructed players per match). The second was an enriched dataset, in which the matches present in the original base were expanded to include a larger number of opponents, reaching an average of 6 or more reconstructed players per lobby.

2. **Covariate dimensionality:** On each dataset, one version was trained with the full set of constructed features (full model, with more than 51 covariates) and another with reduced dimensionality. For the latter, Principal Component Analysis (PCA) was applied to identify the most informative variables, condensing them into a total of 25 variables without losing the fundamental explained variance of the data.

To rigorously evaluate the performance of these four variants on the tournament test data, a set of metrics was used comprising the Integrated Brier Score, Log-loss, Spearman’s correlation, Kendall’s Tau coefficient, and the Concordance Index (C-index).

### 3.3 Model Comparison on the Test Set

Table 1 summarizes the metrics obtained by the four BART models on the tournament test data, evaluated in terms of probabilistic calibration (Integrated Brier Score and mean Log-loss) and discriminative ability (C-index, Spearman’s  $\rho$ , and Kendall’s  $\tau$ ).

Table 1: Predictive performance metrics of the four BART model variants on the tournament test set. For IBS and Log-loss, a lower value indicates better calibration; for C-index,  $\rho$ , and  $\tau$ , a higher value indicates better discriminative ability. The best value per column is highlighted in **bold**.

| Model                         | Dataset / Dim.        | IBS ↓        | Log-loss ↓   | C-index ↑    | $\rho$ ↑     | $\tau$ ↑     |
|-------------------------------|-----------------------|--------------|--------------|--------------|--------------|--------------|
| <code>model_full</code>       | Original / Full (51+) | 0.111        | 0.353        | <b>0.630</b> | <b>0.350</b> | <b>0.345</b> |
| <code>model_reduced</code>    | Original / PCA (25)   | 0.105        | 0.337        | 0.611        | 0.295        | 0.278        |
| <code>model_6p_reduced</code> | Enriched / PCA (25)   | 0.099        | 0.320        | 0.500        | 0.039        | 0.030        |
| <code>model_6p_full</code>    | Enriched / Full       | <b>0.097</b> | <b>0.315</b> | 0.472        | −0.043       | −0.053       |

From the review of the metrics, it can be observed that the models trained on the enriched dataset (`model_6p_*`) obtain the best IBS and Log-loss values, but the worst in discrimination, with a C-index around or below chance (0.500 and 0.472) and rank-correlation coefficients that are practically null or negative. In contrast, the models trained on the original dataset recover ordering ability, with `model_full` being the one that simultaneously maximizes the three concordance estimators ( $C = 0.630$ ,  $\rho = 0.350$ ,  $\tau = 0.345$ ).

To select among the four variants under a single criterion, each metric was normalized to the range  $[0, 1]$ , orienting all scales in the same direction (zero indicates the best relative performance), and they were aggregated into a composite score. Table 2 presents the results.

The selected model was `model_full`, trained on the original dataset and with the full set of constructed covariates (without PCA reduction), with a composite score of 0.394. It is worth noting that the difference relative to the next two variants (`model_reduced` at 0.400 and `model_6p_reduced` at 0.404) is marginal, which evidences a real trade-off

Table 2: Normalized metrics and composite selection score. All scales were oriented so that 0 corresponds to the best relative performance and 1 to the worst. The composite score aggregates the five normalized metrics.

| Model                         | $z_{\text{IBS}}$ | $z_{\text{Log-loss}}$ | $z_{\text{C-index}}$ | $z_{\rho}$ | $z_{\tau}$ | Composite Score ↓ |
|-------------------------------|------------------|-----------------------|----------------------|------------|------------|-------------------|
| <code>model_full</code>       | 0.126            | 0.117                 | 0.824                | 0.790      | 0.793      | <b>0.394</b>      |
| <code>model_reduced</code>    | 0.000            | 0.000                 | 1.000                | 1.000      | 1.000      | 0.400             |
| <code>model_6p_reduced</code> | 0.576            | 0.581                 | 0.118                | 0.140      | 0.168      | 0.404             |
| <code>model_6p_full</code>    | 1.000            | 1.000                 | 0.000                | 0.000      | 0.000      | 0.600             |

between calibration and discrimination: the enriched models produce better-calibrated absolute probabilities but lose the ability to correctly order players within the lobby. Since the substantive objective of this work is to predict the order of elimination, the concordance metrics implicitly received greater weight in the decision, in line with the criterion used by the proposed composite score.

### 3.4 Survival Curves and Intervals of the Winning Model

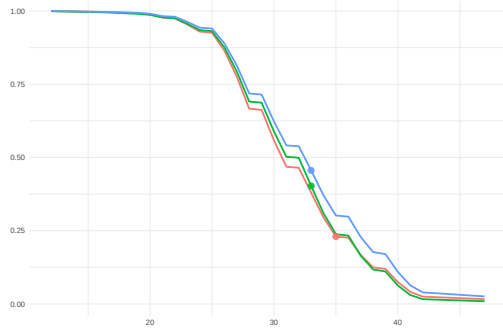
Figures 1 and 2 present the probabilistic estimates obtained by the selected model (`model_full`) over the nine tournament lobbies. Figure 1 shows the posterior probability of remaining alive as a function of the round for each reconstructed player. Figure 2 summarizes the posterior distribution of the elimination round through its 90 % and 95 % credible intervals, together with the predicted median and the actual observed round.

## 4 Discussion

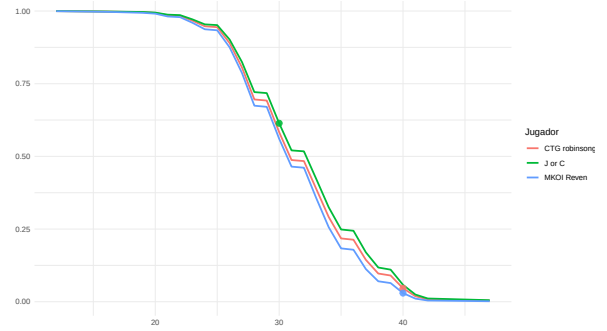
### 4.1 Predictive Performance on Tournament Data

When comparing the performance of the four architectures, the winning model was the one trained on the original dataset using all of its components (without PCA reduction).

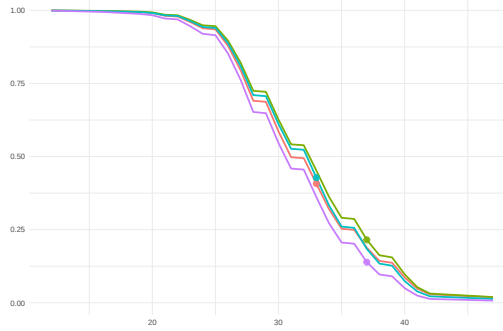
The analysis of this result, beyond what the statistical tests show, revealed that the advantage of the original model (`model_full`) over the enriched models (`model_6p`) was due to the structure of the evaluation data. Because of the inherent limitations in extracting information about all tournament participants, the reconstructed test lobbies predominantly contained between 2 and 4 players. Consequently, the original model, which had been trained under conditions of similar information scarcity (partial lobbies), generalized and adapted to the evaluation environment much better than the enriched model, which expected a more densely populated competitive context (6 or more players). Likewise, the version without dimensional reduction outperformed the PCA version, suggesting that BART is able to efficiently handle high dimensionality and extract value from the nonlinear interactions among the more than 40 original covariates without overfitting.



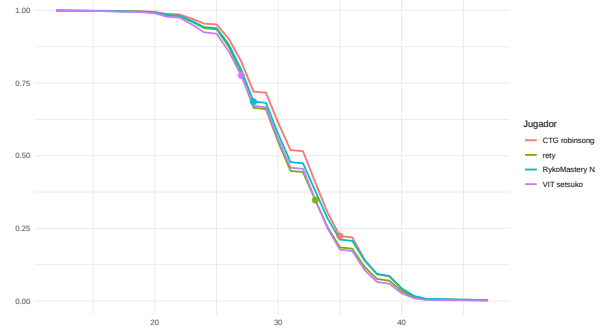
(a) custom\_lobby\_01



(b) custom\_lobby\_04



(c) custom\_lobby\_05



(d) custom\_lobby\_06

Figure 1: Posterior survival curves estimated by the `model_full` model over the reconstructed tournament lobbies with three or four players. The  $x$ -axis represents the game round and the  $y$ -axis the posterior probability of remaining alive.

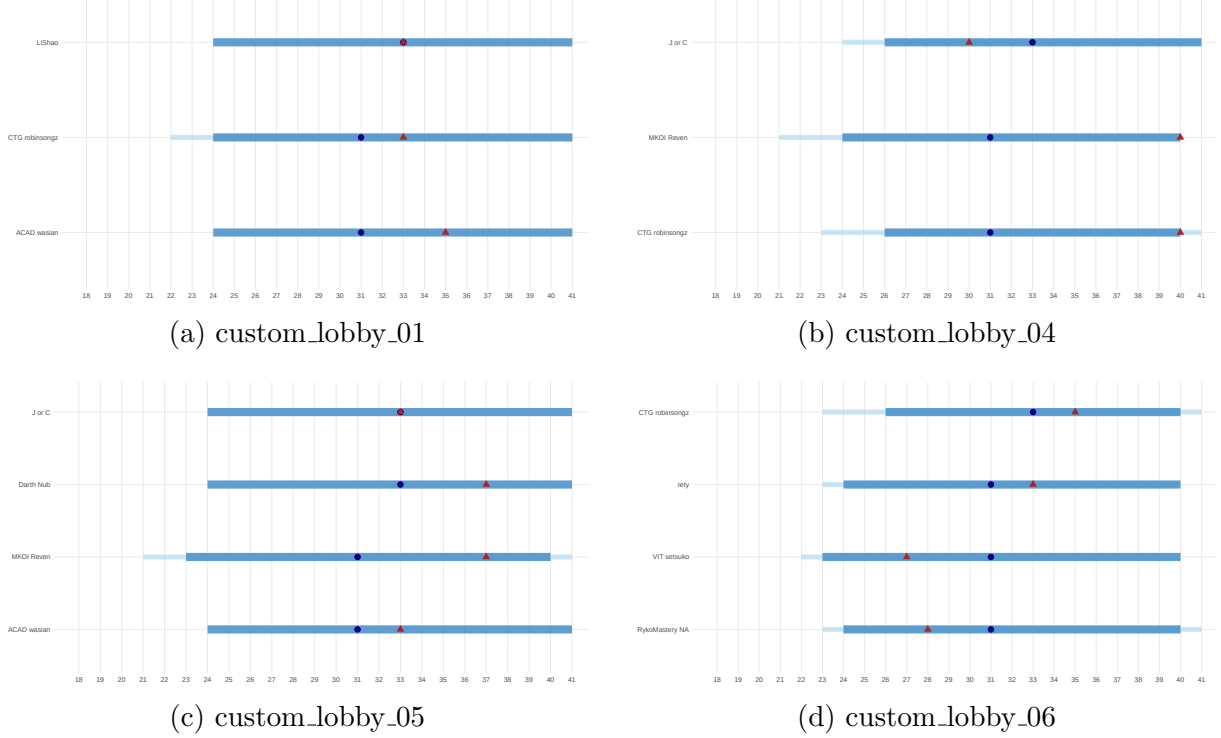


Figure 2: Predictive elimination intervals at 90 % (dark blue) and 95 % (light blue) estimated by the `model_full` model. The blue dot indicates the predicted median of the elimination round and the red triangle the actual observed round.

## 4.2 Analysis of Survival Curves and Credible Intervals

The probabilistic estimates produced by the winning model showed remarkable coherence with the empirical macro-dynamics of the matches. When visually analyzing the survival curves estimated by the BART algorithm, the model was observed to accurately reflect the staged behavior expected in a TFT match. Specifically, the survival function projects moments of stabilization with minimal changes, which empirically coincide with the rounds where there are no confrontations between players (such as PvE rounds or carousels). The curves also partially reflect the order in which players are eliminated: profiles with early drops in their predictive probability corresponded to the first to be eliminated, whereas those assigned sustained survival fell in the late stages of the match.

Additionally, the analysis of predictive uncertainty yielded mixed results. On the one hand, when contrasting the actual elimination round against the estimates, all empirical observations were found to fall successfully within the credible intervals generated by the MCMC iterations (at 90% and 95%). For 16 of the 24 tournament participants simulated within their respective lobbies, the distance from the predicted median to the actual value was notably close, lying around 3 rounds or less of difference, rising to 20 of 24 if we take a difference of 4 or fewer rounds.

On the other hand, the generated credible interval was found to be too wide to offer predictions with meaningful precision. Although on average the central errors are moderate, in several cases the actual value lay considerably far from the mean, falling at the extreme limits of the credible interval. In practice, the interval comes to cover a range

exceeding 2 or more game stages (from 18 to 21 rounds). This extreme width compromises the pointwise usefulness of the prediction, suggesting a high degree of dispersion in the estimates.

### 4.3 Analysis and Interpretation

Statistical modeling in *Battle Royale* or zero-sum competitions assumes high interdependence among observations. This work shows that individual prediction errors can become conditionally exchangeable if one correctly parameterizes the recent competitive history and the relative performance advantages against the available lobby.

A central aspect of the obtained results is the methodology used in the design of the *pipeline*: the model issues its predictions using exclusively pre-match information. The fact that the survival curves manage to capture the structure of the match (such as the pauses in rounds without confrontations) and to empirically order the eliminations based solely on smoothed past behavior validates the soundness of the extracted covariates.

Nevertheless, the empirical evaluation exposes two fundamental limitations. The first is related to the collection of competitive context. By operating with partially reconstructed lobbies, the relative-strength metrics were computed against subsets of opponents and not over the 8 real participants, which prevented exploiting the full potential of the enriched model.

The second structural limitation lies in the precision capacity of the algorithm. Although the observations fall within the credible intervals and there are cases of high accuracy (with distances of only 3 rounds), the total width of the interval is excessive. This excessive width indicates that, while pre-match historical information is robust for projecting relative trends and the merit ordering of the lobby, it is insufficient to isolate the exact moment of elimination. The competitive environment of TFT possesses an extreme inherent variance, governed by elements observed only during the course of the game, such as shop randomness, matchmaking, and immediate tactical decisions, which makes it impossible to reduce the uncertainty to a narrow predictive window using only the players' past.

Despite this dispersion, the main theoretical advantage of Bayesian modeling has been demonstrated: BART naturally learned the non-proportional hazards and discovered the complex functional geometry that links experience and historical consistency without requiring restrictive functional forms.

## 5 Reproducible Workflow

To guarantee the traceability of the study and to ensure that the statistical architecture integrates correctly with the data from the real competitive environment, a processing and modeling pipeline was designed and implemented, structured in the following sequential stages:

1. **Account Resolution and Match Extraction:** Identification of players through the Riot Games API (resolving Riot IDs to PUUIDs) and retrieval of valid ranked matches, applying the level and minimum-round thresholds to ensure the maturity of the cohort.

2. **Per-Player History Construction:** Dynamic computation of each subject’s past performance metrics. This step strictly guarantees that, for any target row, only information chronologically available before the start of that match is computed.
3. **Stabilization via Burn-in and Smoothing:** Application of an initial data-accumulation period (10 matches per player) during which no target predictions are issued, allowing their metrics to be stabilized through empirical-Bayes shrinkage that pulls the initial averages toward the global mean.
4. **Lobby Reconstruction and Relative Context:** Grouping of the players matched in the same game to compute relative metrics against the available opponents (positioning advantages, consistency, and win rate).
5. **Longitudinal Expansion:** Transformation of the database from a cross-sectional format into a discrete survival matrix. Each match is unfolded into a temporal binary sequence indexed by the round reached (`last_round`).
6. **BART Model Training:** Fitting of the Bayesian ensemble using the latent-variable approach with a probit link to relate the historical and lobby covariates to the survival probability at each round.
7. **Prediction and Evaluation:** Generation of dynamic probabilities and comparison against the actual observed placements and rounds, applying a fixed predictive-horizon strategy to ensure that the evaluation does not suffer from future-data contamination.

## 6 References

### References

- [1] Sparapani, R. A., Logan, B. R., McCulloch, R. E., & Laud, P. W. (2016). Nonparametric survival analysis using Bayesian Additive Regression Trees (BART). *Statistics in Medicine*, 35(16), 2741–2753.
- [2] Albert, J., & Chib, S. (1993). Bayesian analysis of binary and polychotomous response data. *Journal of the American Statistical Association*, 88(422), 669–679.
- [3] Chipman, H. A., George, E. I., & McCulloch, R. E. (2010). BART: Bayesian Additive Regression Trees. *The Annals of Applied Statistics*, 4(1), 266–298.
- [4] Cox, D. R. (1972). Regression models and life tables. *Journal of the Royal Statistical Society. Series B (Methodological)*, 34(2), 187–220.
- [5] Riot Games. (2023). Riot Games Developer API. Teamfight Tactics Documentation. URL: <https://developer.riotgames.com/>